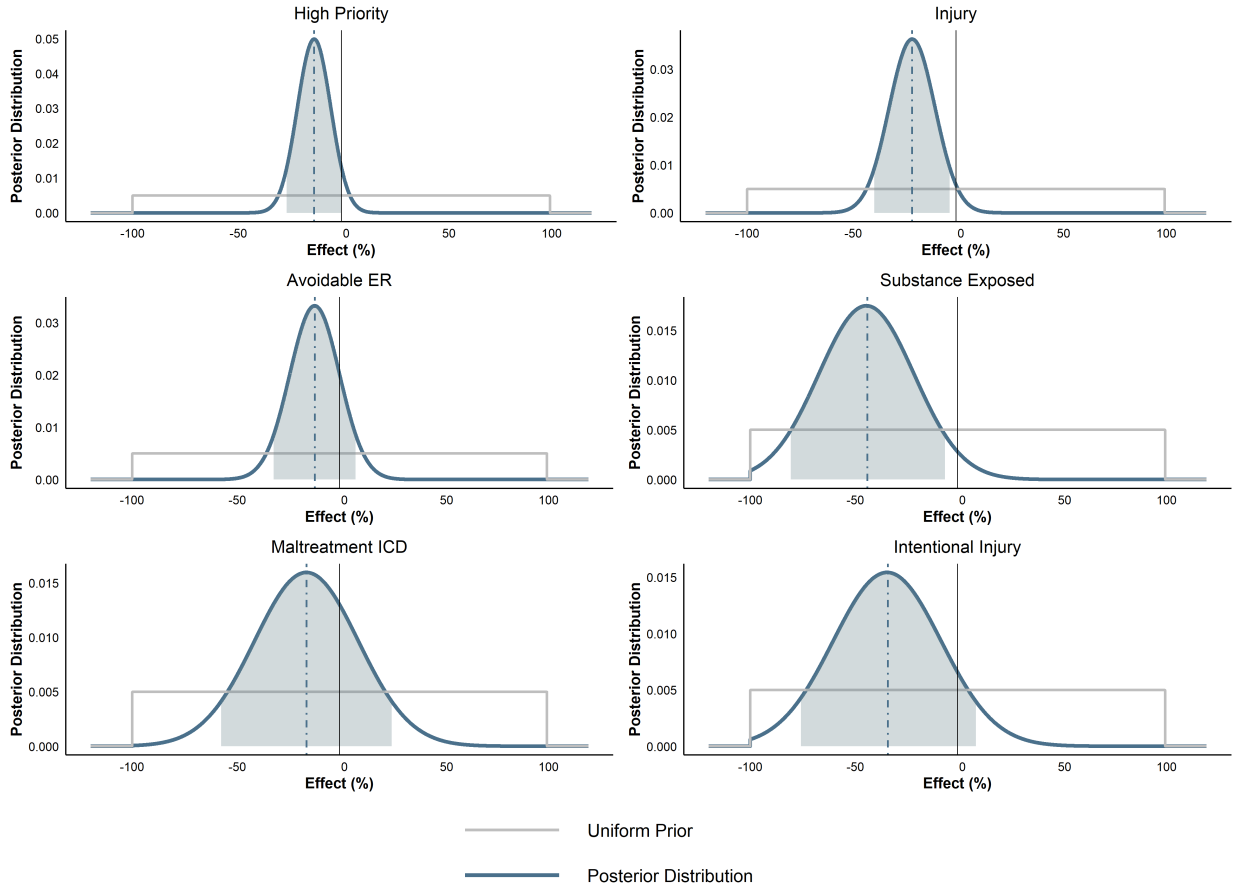
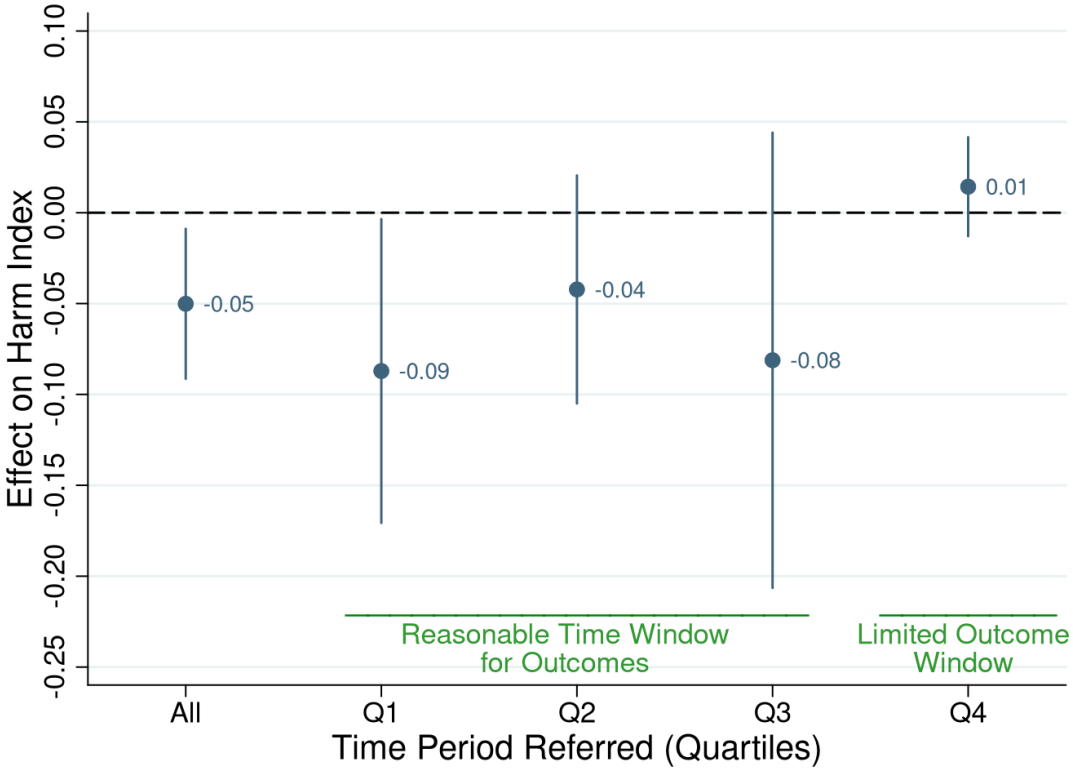
Figure A.6: Bayesian Posterior Distribution of Effects on Harm Index Components



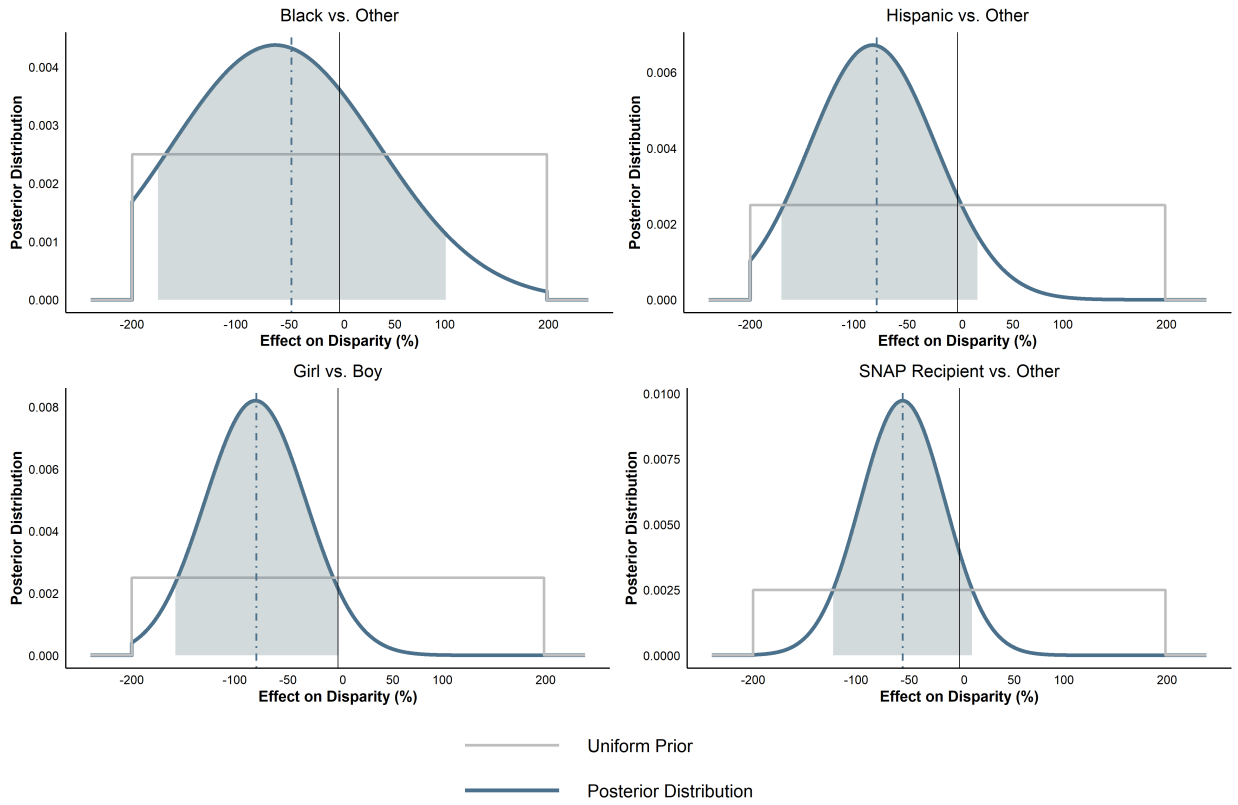
Notes: These figures report the posterior distribution of the effect of algorithm support on six components of the harm index: counts of high priority visits, injury visits, avoidable ER visits, substance exposure incidents, visits with explicit maltreatment codes (rare), and visits with intentional injuries. We assume as an uninformative uniform prior that effects could have been anywhere in the range of -100 percent to +100 percent (gray line centered around vertical line at 0). Following Brännlund et al. (2024), we then calculate the posterior distribution given the data we observe (posterior distribution outlined in blue; observed mean marked by dash-dot vertical blue line). 90% credible intervals are reported as the shaded areas.

Figure A.7: Estimated Effects by Time of First Appearance in the Trial



Notes: This figure presents estimates of the tool on harm by the time a child was first seen during the trial. On the left, we present the main (full sample) point estimate of providing the algorithm tool on child harm ("All"). The full sample is then split into four equally sized groups by date of referral and estimate the effects of providing the tool on the harm index in each of these sample quarters. Q1 is the first quarter of children referred during the trial, Q2 the second, and so on. 90% confidence intervals are reported throughout. Absence of impacts for children referred in the final quarter (Figure A.7) could be due to a limited window of time to observe hospital outcomes for this subsample during the trial. After the end of the trial, the algorithm tool became available for all children being referred, attenuating effects.

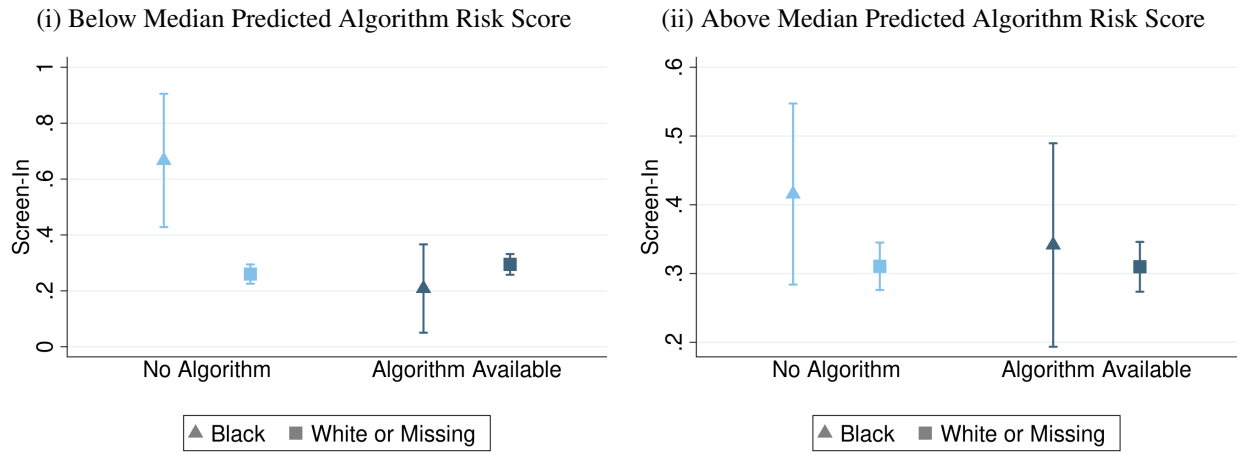
Figure A.8: Bayesian Posterior Distribution of Effects on Disparities with an Uninformative Prior



Notes: These figures report the posterior distribution of the effect of algorithm support on four harm index disparity margins: race, ethnicity, gender and socioeconomic status. We assume as an uninformative uniform prior that effects could have been anywhere in the range of -200% to +200% (gray line centered around vertical line at 0). Given the data and following Brännlund et al. (2024), we then calculate the posterior distribution given the data we observe (posterior distribution outlined in blue; observed mean marked by dash-dot vertical blue line). 90% credible intervals are reported as the shaded areas.

Figure A.9: Tool Reduced Racial Disparities for Low-Risk-Score Children

Using CPS-Known Race and Hospital-Known Race



Notes: These figures show the estimates of mean harm index by demographics and tool availability. Figures define race as ever being recorded as Black by Child Protective Services or hospitals. 90% confidence intervals are reported and randomization controls are included throughout.